



Digital Signal Processing

Lecture 7 Z Transform

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z-Transform

Background-Definition

- Fourier transform $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$ extracts the essence of $x[n]$

but is limited in the sense that it can handle stable systems only.

$$X(e^{j\omega}) \text{ converges if } \sum |x[n]| < \infty$$

i.e., stable system $\rightarrow$ Fourier Transform converges

- So, we want to extend it such that it can be used as a tool to analyze digital systems in general.

Let $X_r(e^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n}$

then it converges if $\sum |x[n]r^{-n}| < \infty$

The condition for convergence is relaxed!

(e.g.) $x[n] = 2^n u[n]$

$$|X(e^{j\omega})| = \left| \sum 2^n e^{-j\omega n} \right| \rightarrow \infty$$

$$|X_r(e^{j\omega})| = \left| \sum 2^n r^{-n} e^{-j\omega n} \right| < \sum 2^n r^{-n} = \frac{1}{1 - \frac{2}{|r|}}$$

$\rightarrow$ converges if $|r| > 2$

- This implies that $X_r(e^{j\omega})$ can handle some systems that $X(e^{j\omega})$ cannot due to divergence.
- Therefore we define z-transform to be

$$X(z) = X_r(e^{j\omega})|_{re^{j\omega}=z} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Representing the condition for convergence of $X_r(e^{j\omega})$ in terms of region of convergence RoC.

(e.g.) in case $x[n] = 2^n u[n]$

$X_r(e^{j\omega})$ exists for $|r| > 2$.

So, RoC is $|z| = |re^{j\omega}| > 2$.

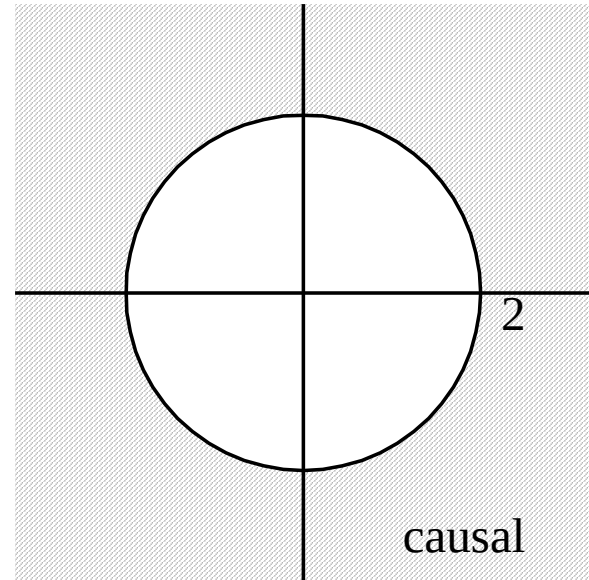
In general, if $x[n] = a^n u[n]$

$R_o C$ is $|z| > |a|$

- In terms of $X(z)$,

$X(e^{j\omega})$ is a special case

Where $|z| = 1$, or $r = 1$



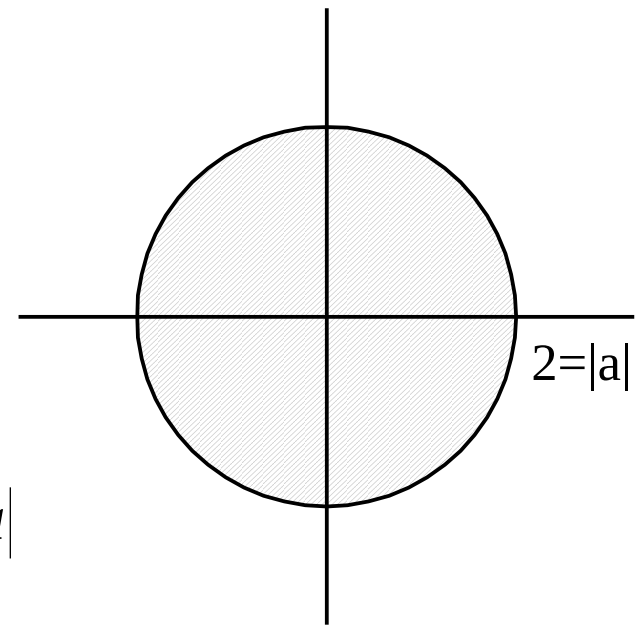
(e.g.) $x[n] = -a^n u[-n-1]$

$$X(z) = - \sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n} = - \sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= - \sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n$$

$$= 1 - \frac{1}{1 - a^{-1} z} = \frac{-a^{-1} z}{1 - a^{-1} z}$$

$$= \frac{1}{1 - a z^{-1}}$$



$$R_o C \quad : \quad |a^{-1} z| < 1, \text{ or } |z| < |a|$$

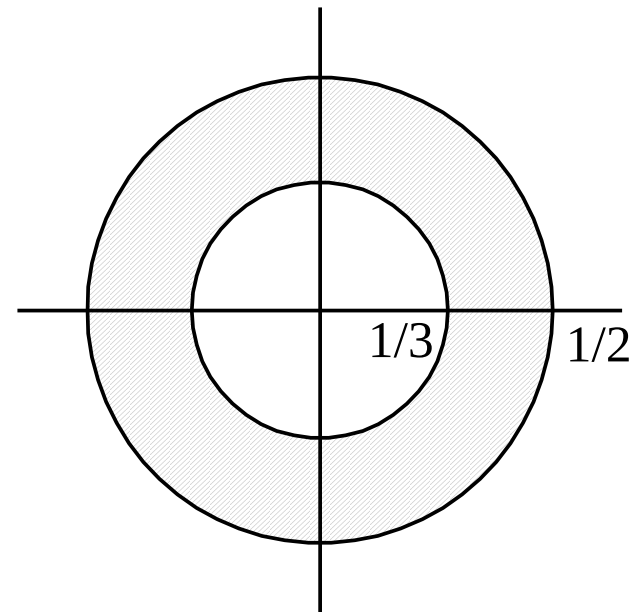
(e.g.) Two - sided sequence

$$x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$$

$$X(z) = \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n z^{-n} - \sum_{n=-\infty}^{-1} \left(\frac{1}{2}\right)^n z^{-n}$$

$$= \frac{1}{1 + \frac{1}{3}z^{-1}} - \frac{1}{1 - \frac{1}{2}z^{-1}}$$

$$|z| > \frac{1}{3}, \quad |z| < \frac{1}{2}$$



Some Common z-Transforms

$$(1) \quad \delta[n] \quad \leftrightarrow \quad 1 \quad [all \ z]$$

$$(2) \quad u[n] \quad \leftrightarrow \quad \frac{1}{1 - z^{-1}} \quad [|z| > 1]$$

$$(3) \quad -u[-n-1] \quad \leftrightarrow \quad \frac{1}{1 - z^{-1}} \quad [|z| < 1]$$

$$(4) \quad \delta[n-m] \quad \leftrightarrow \quad z^{-m} \quad [all \ z \ except \ 0, \ if \ m > 0, \\ all \ z \ except \ \infty, \ if \ m < 0]$$

$$(5) \quad a^n u[n] \quad \leftrightarrow \quad \frac{1}{1 - az^{-1}} \quad [|z| > |a|]$$

$$(6) \quad -a^n u[-n-1] \quad \leftrightarrow \quad \frac{1}{1 - az^{-1}} \quad [|z| < |a|]$$

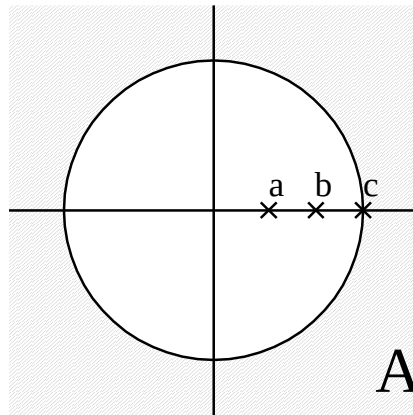
$$(7) \quad na^n u[n] \quad \leftrightarrow \quad \frac{az^{-1}}{(1 - az^{-1})^2} \quad [|z| > |a|]$$

Properties of R_oC

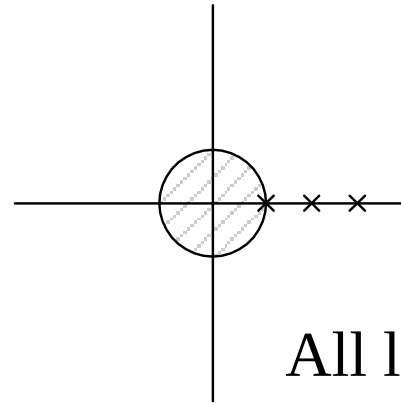
- (1) in general $0 \leq r_R < R_oC < r_L \leq \infty$
- (2) $X(e^{j\omega})$ absolutely converges $\leftrightarrow UC \subset R_oC$
- (3) R_oC cannot contain a pole
- (4) FIR sequence $\rightarrow$ entire z plane, may be except for 0 or ∞
- (5) Right-sided sequence $\rightarrow$ outward of the outermost pole
- (6) Left-sided sequence $\rightarrow$ inward from the innermost pole
- (7) Two-sided sequence $\rightarrow$ a ring in between two adjacent rings
- (8) R_oC is a connected region

(e.g.) If $x[n]$ is a sum of 3 sequences whose poles are a, b, c respectively,

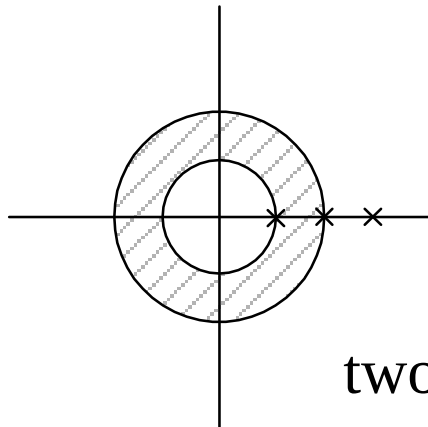
There exist 4 possible ROCs as shown below



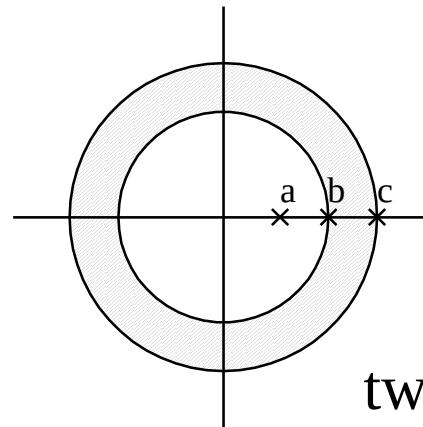
All right-sided



All left-sided



two left-sided



two right-sided

z-Transform Properties

(1) Linearity

$$ax_1[n] + bx_2[n] \quad \leftrightarrow \quad aX_1(z) + bX_2(z)$$

(2) Time shifting

$$x[n - n_o] \quad \leftrightarrow \quad z^{-n_o} X(z)$$

(e.g.) $\frac{z^{-1}}{1 - \frac{1}{4}z^{-1}} \leftrightarrow z^{-1} \cdot \frac{1}{1 - \frac{1}{4}z^{-1}} \rightarrow \left(\frac{1}{4}\right)^n u[n] \quad \text{and delay -1}$

$\rightarrow \left(\frac{1}{4}\right)^{n-1} u[n-1], \quad |z| > \frac{1}{4}$

z-Transform Properties..(cont.)

(3) Multiplication by an Exponential Sequence

$$z_o^n x[n] \leftrightarrow X\left(\frac{z}{z_o}\right)$$

$$\text{(e.g.) } (e^{jw_o})^n x[n] \leftrightarrow X(e^{-jw_o} \cdot z) \quad \begin{matrix} z = e^{jw} \\ \rightarrow \end{matrix} \quad X(e^{j(w-w_o)})$$

$$\begin{aligned} \text{(e.g.) } r^n \cos w_o n \cdot u[n] &= \frac{r^n}{2} [(e^{jw_o})^n + (e^{-jw_o})^n] u[n] \\ &= \frac{1}{2} [(re^{jw_o})^n + (re^{-jw_o})^n] u[n] \\ &\leftrightarrow \frac{1}{2} \left[\frac{1}{1 - re^{jw_o} \cdot z^{-1}} + \frac{1}{1 - re^{-jw_o} z^{-1}} \right] \\ &= \frac{1 - r \cos w_o z^{-1}}{1 - 2r \cos w_o z^{-1} + r^2 z^{-2}} \quad |z| > r \end{aligned}$$

z-Transform Properties..(cont.)

(4) Differentiation of $X(z)$

$$nx[n] \leftrightarrow -z \frac{d}{dz} x[z] \qquad R_o C = R_x$$

(e.g.) $X(z) = \log(1 + az^{-1}) \qquad |z| > |a|$

$$\frac{dX(z)}{dz} = \frac{-az^{-2}}{1 + az^{-1}}$$

$$-z \frac{d}{dz} X(z) = \frac{az^{-1}}{1 + az^{-1}} = a \cdot z^{-1} \cdot \frac{1}{1 + az^{-1}}$$

$$nx[n] = a(-a)^{n-1} u[n-1]$$

$$x[n] = (-1)^{n-1} \frac{a^n}{n} u[n-1]$$

z-Transform Properties..(cont.)

(5) Conjugation of Complex Sequence

$$x^*[n] \leftrightarrow X^*(z^*)$$

$$R_o C = R_x$$

(6) Time-Reversal

$$x^*[-n] \leftrightarrow X^*\left(\frac{1}{z^*}\right)$$

$$R_o C = \frac{1}{R_x}$$

$$x[-n] \leftrightarrow X\left(\frac{1}{z}\right)$$

(7) Convolution-Integration

$$x_1[n] * x_2[n] \leftrightarrow X_1(z)X_2(z)$$

Inverse z-Transform

4-Ways:

- By the contour integral of the inverse transform definition
- “By Inspection” - recognize common transform pairs
- Partial Fraction Expansion - effective for rational z-transforms
- Power Series Expansion

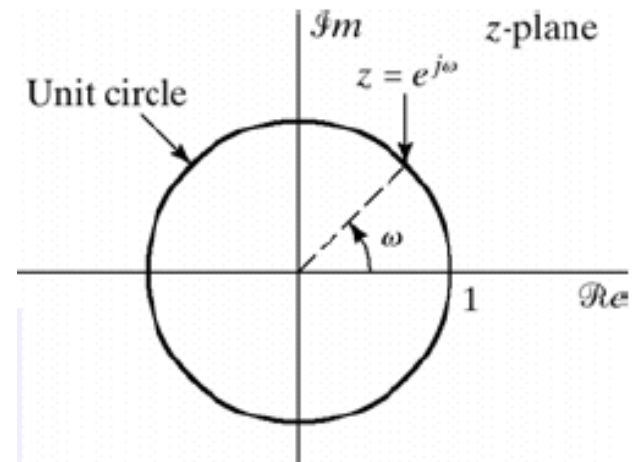
Inversion by Contour Integration

- Cauchy integral definition of the inverse z-Transform

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

- Example: Inverse DTFT
- Implies.. Contour C is chosen as unit circle $z = e^{j\omega}$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$



Inversion Method -1

Concept of Partial Fraction Expansion-

- Consider a general rational z -transform (no repeated roots); two equivalent forms are:

$$X(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}$$

Inversion Method

Concept of Partial Fraction Expansion Inversion-

1. Find partial fraction expansion method in third equivalent form

$$X(z) = \underbrace{\left[\sum_{r=0}^{(M-N)} B_r z^{-r} \right]}_{\text{only if } M \geq N} + \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}}$$

2. Invert by expansion

4. $\delta[n - m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $a^n u[n]$	$\frac{1}{1 - a z^{-1}}$	$ z > a $
6. $-a^n u[-n - 1]$	$\frac{1}{1 - a z^{-1}}$	$ z < a $

Doing the Partial Fraction Expansion

To find the coefficients of the m^{th} first-order pole, consider

$$(1 - d_m z^{-1})X(z) = (1 - d_m z^{-1}) \left[\sum_{r=0}^{(M-N)} B_r z^{-r} \right] + \sum_{k=1}^N \frac{(1 - d_m z^{-1})A_k}{1 - d_k z^{-1}}$$

Now evaluate at $z = d_m \dots$

Doing the Partial Fraction Expansion-2

Thus we now have:

$$(1 - d_m z^{-1})X(z) \Big|_{z=d_m} = (1 - \cancel{d_m} z^{-1} \overset{0}{\nearrow}) \left[\sum_{r=0}^{(M-N)} B_r z^{-r} \right] \Big|_z \\ + \sum_{k=1}^N \frac{(1 - \cancel{d_m} z^{-1} \overset{0}{\nearrow}) A_k}{1 - d_k z^{-1}} \Big|_{z=d_m} = A_m$$

$$A_k = (1 - d_k z^{-1})X(z) \Big|_{z=d_k}$$

We will see that this formula is applied to the product form in practice ...

Writing Down $x[n]$

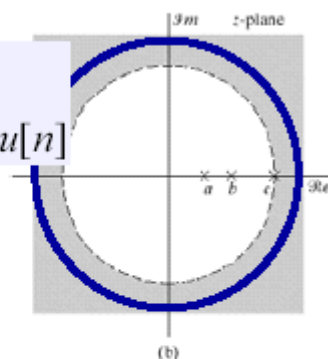
$$X(z) = \underbrace{\left[\sum_{r=0}^{(M-N)} B_r z^{-r} \right]}_{\text{if } M \geq N} + \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}}$$

$$x[n] = \underbrace{\left[\sum_{r=0}^{(M-N)} B_r \delta[n-r] \right]}_{\text{if } M \geq N} + \underbrace{\sum_k A_k d_k^n u[n]}_{\text{when } |d_k| < r_R} - \underbrace{\sum_k A_k d_k^n u[-n-1]}_{\text{when } |d_k| > r_L}$$

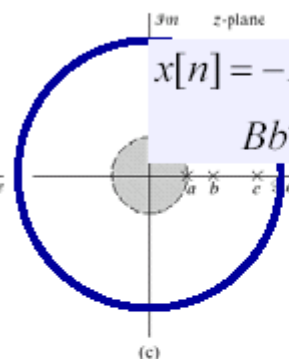
$X[n]$ depend on knowing the ROC

$$X(z) = \frac{A}{1 - az^{-1}} + \frac{B}{1 - bz^{-1}} + \frac{C}{1 - cz^{-1}}$$

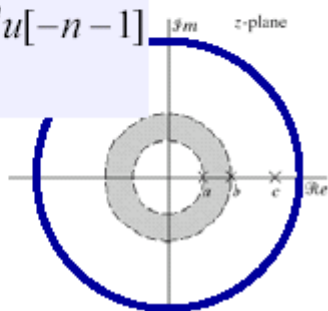
$$x[n] = Aa^n u[n] + Bb^n u[n] + Cc^n u[n]$$



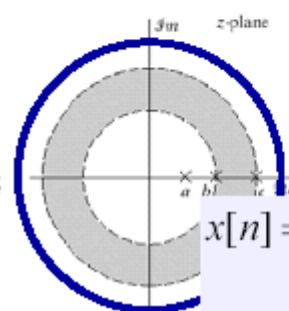
$$x[n] = -Aa^n u[-n-1] - Bb^n u[-n-1] - Cc^n u[-n-1]$$



$$x[n] = Aa^n u[n] - Bb^n u[-n-1] - Cc^n u[-n-1]$$



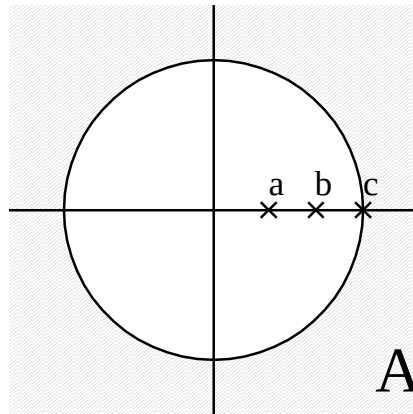
$$x[n] = Aa^n u[n] + Bb^n u[n] - Ca^n u[-n-1]$$



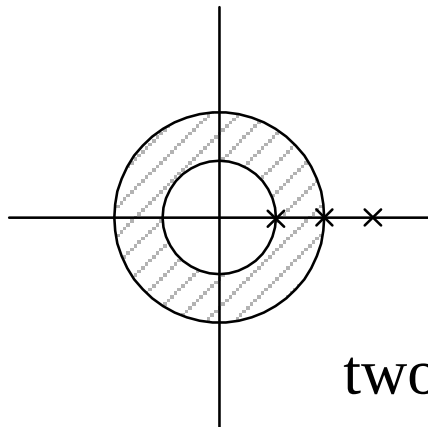
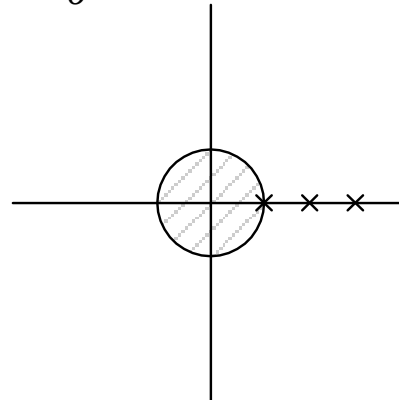
Example- ROC

If $x[n]$ is a sum of 3 sequences whose poles are a, b, c respectively,

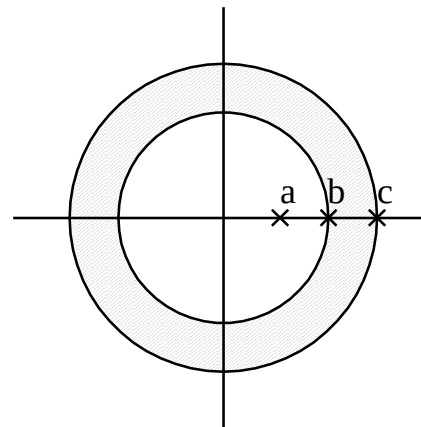
There exist A possible R_oCs as shown below



All right-sided



two left-sided



Example- Partial Fraction

$$X(z) = \frac{1 + 3z^{-1} + 3z^{-2} + z^{-3}}{1 - 3z^{-1} + 2z^{-2}} = \frac{(1 + z^{-1})^3}{(1 - z^{-1})(1 - 2z^{-1})}$$

Long Division

$$X(z) = \frac{1 + 3z^{-1} + 3z^{-2} + z^{-3}}{1 - 3z^{-1} + 2z^{-2}} = \frac{(1 + z^{-1})^3}{(1 - z^{-1})(1 - 2z^{-1})}$$

$$0.5z^{-1} + 2.25$$

$$\begin{array}{r} \underline{2z^{-2} - 3z^{-1} + 1} \quad \boxed{\begin{array}{r} z^{-3} + 3.0z^{-2} + 3.0z^{-1} + 1.0 \\ \underline{z^{-3} - 1.5z^{-2} + 0.5z^{-1}} \\ 4.5z^{-2} + 2.50z^{-1} \\ \underline{4.5z^{-2} - 6.75z^{-1} + 2.25} \\ 9.25z^{-1} - 1.25 \end{array}} \end{array}$$

$$X(z) =$$

$$2.25 + 0.5z^{-1} - \frac{1.25 - 9.25z^{-1}}{1 - 3z^{-1} + 2z^{-2}} = 2.25 + 0.5z^{-1} - \frac{1.25 - 9.25z^{-1}}{(1 - z^{-1})(1 - 2z^{-1})}$$

Finding the coefficients of Poles

$$X(z) = \frac{(1 + z^{-1})^3}{(1 - z^{-1})(1 - 2z^{-1})}$$

$$= 2.25 + 0.5z^{-1} + \frac{A_1}{(1 - z^{-1})} + \frac{A_2}{(1 - 2z^{-1})}$$

$$A_1 = X(z)(1 - z^{-1}) \Big|_{z=1} = \frac{(1 + z^{-1})^3}{(1 - 2z^{-1})} \Big|_{z=1} = \frac{8}{-1} = -8$$

$$A_2 = X(z)(1 - 2z^{-1}) \Big|_{z=2} = \frac{(1 + z^{-1})^3}{(1 - z^{-1})} \Big|_{z=2} = \frac{(3/2)^3}{1/2} = 6.75$$

$$X(z) = 2.25 + 0.5z^{-1} + \frac{-8}{(1 - z^{-1})} + \frac{6.75}{(1 - 2z^{-1})}$$

Writing Down $x[n]$

$$X(z) = 2.25 + 0.5z^{-1} + \frac{-8}{(1 - z^{-1})} + \frac{6.75}{(1 - 2z^{-1})}$$

- If ROC is $2 < |z|$

$$x[n] = 2.25\delta[n] + 0.5\delta[n-1] - 8u[n] + 6.75(2)^n u[n]$$

- If ROC is $1 < |z| < 2$

$$x[n] = 2.25\delta[n] + 0.5\delta[n-1] - 8u[n] - 6.75(2)^n u[-n-1]$$

- If ROC is $|z| < 1$

$$x[n] = 2.25\delta[n] + 0.5\delta[n-1] + 8u[-n-1] - 6.75(2)^n u[-n-1]$$

Partial Fraction Expansion in MATLAB

$$X(z) = \frac{1 + 3z^{-1} + 3z^{-2} + z^{-3}}{1 - 3z^{-1} + 2z^{-2}} = 2.25 + 0.5z^{-1} + \frac{-8}{1 - z^{-1}} + \frac{6.75}{1 - 2z^{-1}}$$

```
>> [r,p,k]=residuez([1,3,3,1],[1,-3,2])  
r =  
    6.750000000000000  
   -8.000000000000000  
p =  
     2  
     1  
k =  
    2.250000000000000    0.500000000000000
```

MATLAB's `residuez` can also go back the other way, and can also handle repeated roots

– Note: '`residuez`' is part of the optional Signal Processing Toolbox

Selected z-Transform Theorems

- The delay or shift property:

$$x[n - n_0] \Leftrightarrow z^{-n_0} X(z)$$

- The convolution property:

$$y[n] = x[n] * h[n] \Leftrightarrow Y(z) = H(z)X(z)$$

- A consequence:

$$y[n] = x[n] * \delta[n - n_0] \Leftrightarrow Y(z) = z^{-n_0} X(z)$$

An IIR System

- Difference equation:

$$y[n] = ay[n-1] + x[n] \Leftrightarrow Y(z) = az^{-1}Y(z) + X(z)$$

- System function:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \quad |z| > |a| \quad \leftarrow \begin{array}{l} \text{ROC if} \\ \text{causal} \end{array}$$

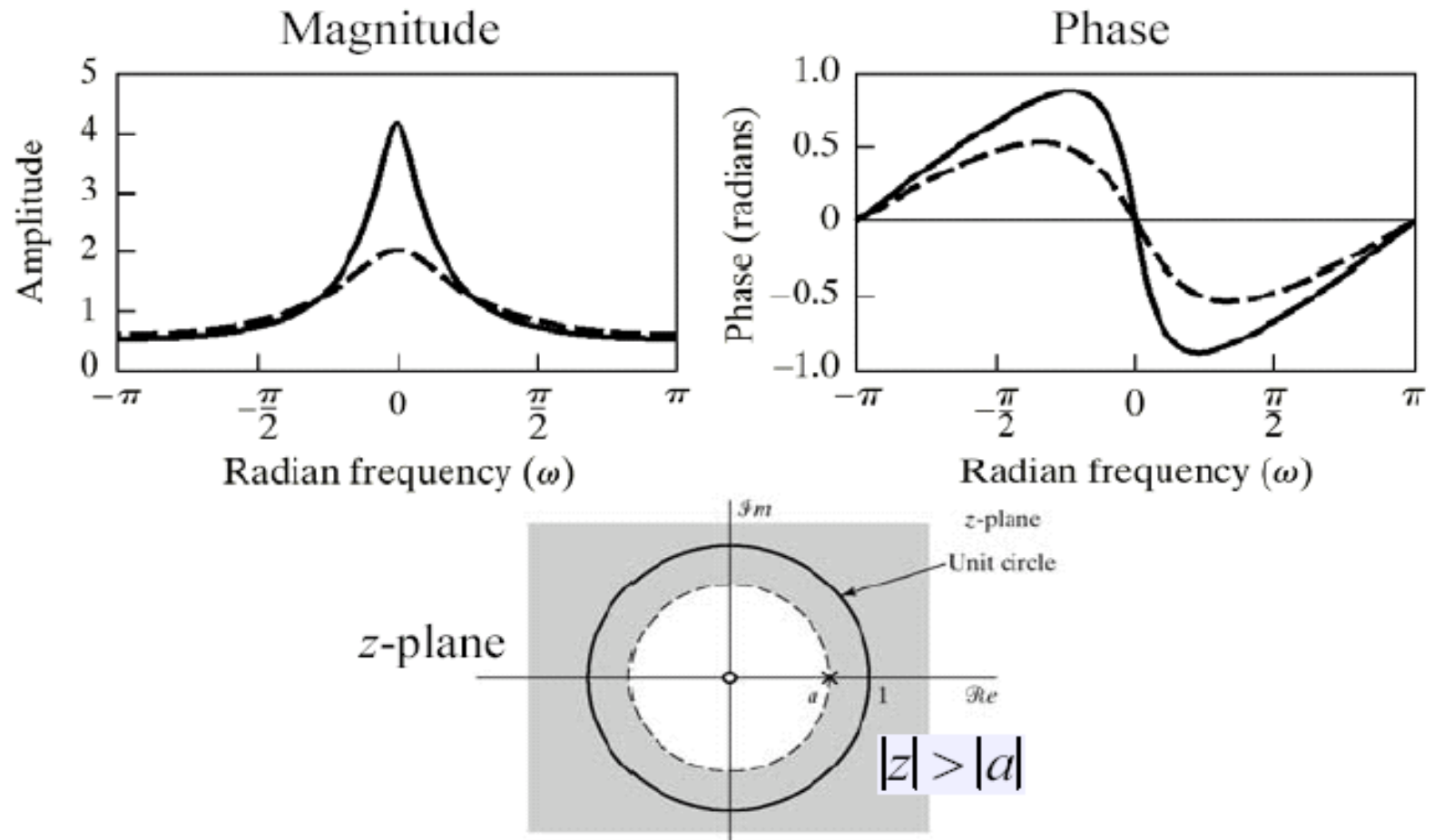
- Frequency response:

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} = \frac{1}{1 - ae^{-j\omega}} \quad \begin{array}{l} \text{Assumes} \\ |a| < 1 \end{array}$$

- Impulse response:

$$h[n] = a^n u[n]$$

IIR Frequency Response



System Function Of a Difference Equation

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$\sum_{k=0}^N a_k Y(z) z^{-k} = \sum_{k=0}^M b_k X(z) z^{-k}$$

$$\left(\sum_{k=0}^N a_k z^{-k} \right) Y(z) = \left(\sum_{k=0}^M b_k z^{-k} \right) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(\sum_{k=0}^M b_k z^{-k} \right)}{\left(\sum_{k=0}^N a_k z^{-k} \right)}$$

Difference equations give rise to rational z transforms!

$H[z]$ and $h[n]$

- Consider a causal system; *i.e.* $h[n]=0$ for $n<0$:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})} \quad \text{ROC: } |z| > \max_k |d_k|$$

$$H(z) = \underbrace{\left[\sum_{r=0}^{(M-N)} B_r z^{-r} \right]}_{\text{only if } M \geq N} + \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}}$$

$$h[n] = \sum_{r=0}^{(M-N)} B_r \delta[n-r] + \sum_{k=1}^N A_k d_k^n u[n]$$

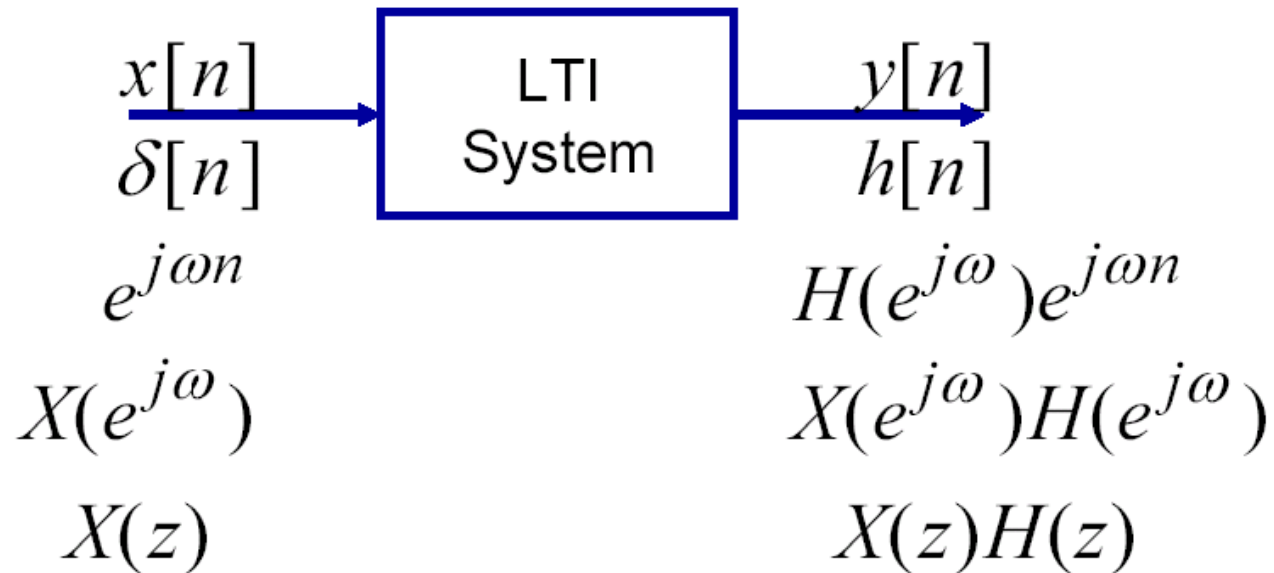
Frequency Response of a DE

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} = \frac{\left(\sum_{k=0}^M b_k e^{-j\omega k} \right)}{\left(\sum_{k=0}^N a_k e^{-j\omega k} \right)}$$

ROC must
Contain the
Unit circle

LTI System Characterization

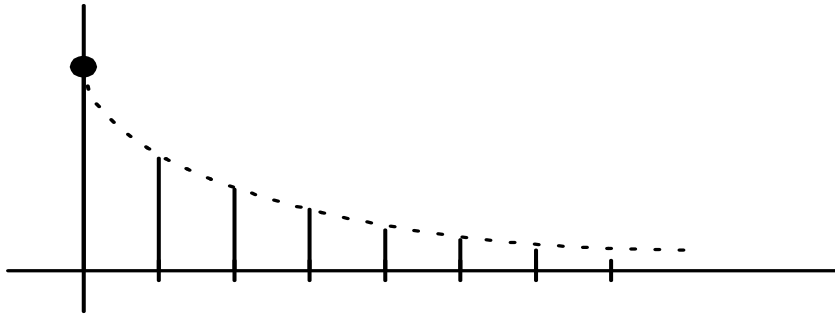


$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(\sum_{k=0}^M b_k z^{-k} \right)}{\left(\sum_{k=0}^N a_k z^{-k} \right)}$$

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$$

Stability, Causality- illustration

(1) $x[n] = \left(\frac{1}{2}\right)^n u[n]$



$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

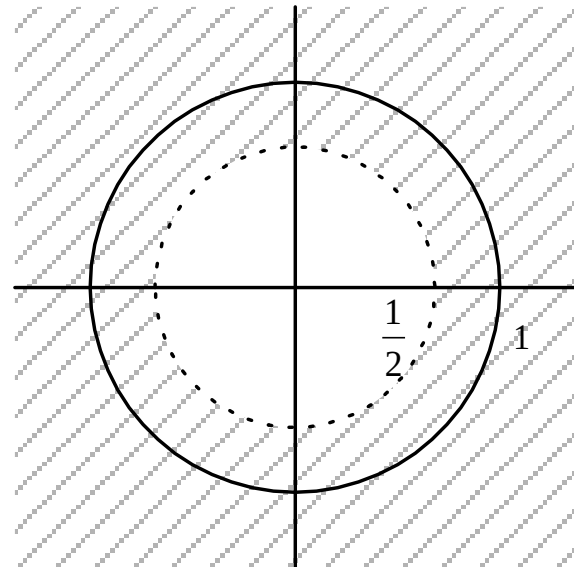
$$\text{RoC} : |z| > \frac{1}{2}$$

① Outward

② $\text{UC} \subset \text{RoC}$

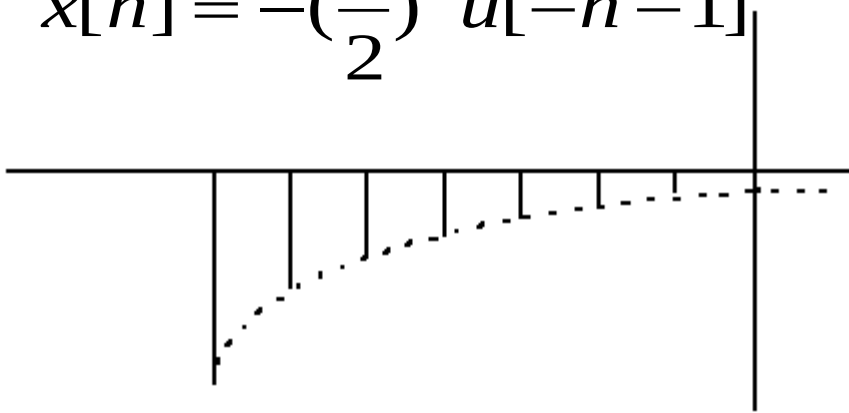
① Causal

② Stable



Stability, Causality- illustration..(cont)

(2) $x[n] = -\left(\frac{1}{2}\right)^n u[-n-1]$



① Anti Causal

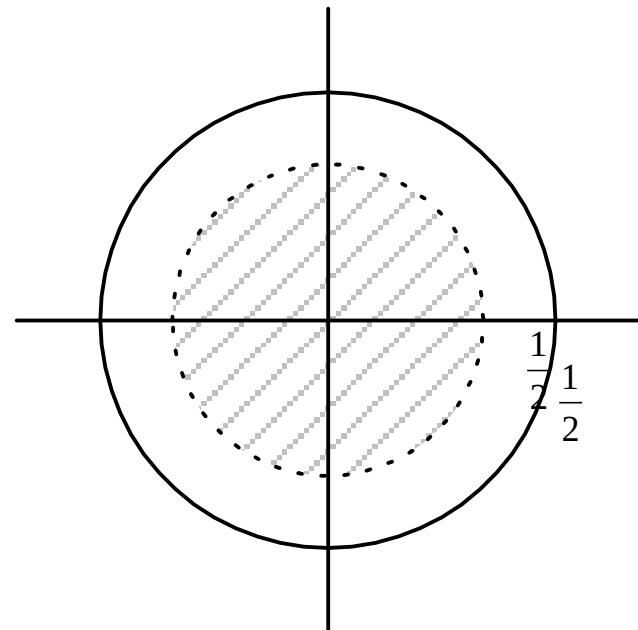
② Unstable

$$X(z) = \sum_{n=-\infty}^{-1} -\left(\frac{1}{2}\right)^n z^{-n} = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

$$\text{ROC} : |z| < \frac{1}{2}$$

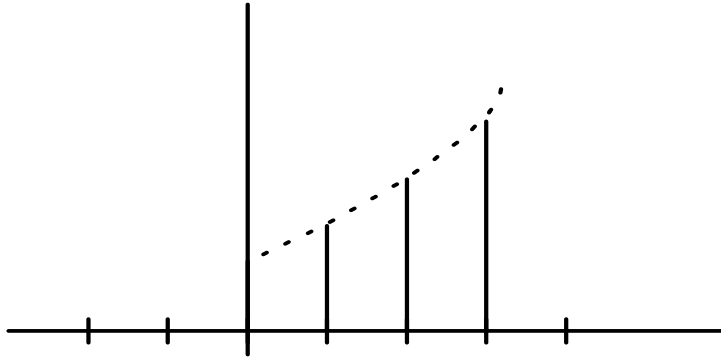
① Inward

② UC $\not\subset$ RoC



Stability, Causality- illustration..(cont)

(3) $x[n] = (2)^n u[n]$



$$X(z) = \sum_{n=0}^{\infty} 2^n z^{-n} = \frac{1}{1-2z^{-1}}$$

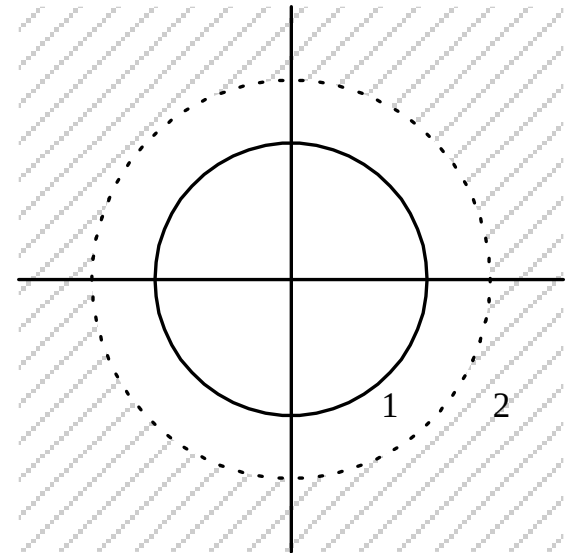
- RoC : $|z| > 2$

① Outward

② ~~UC~~ RoC

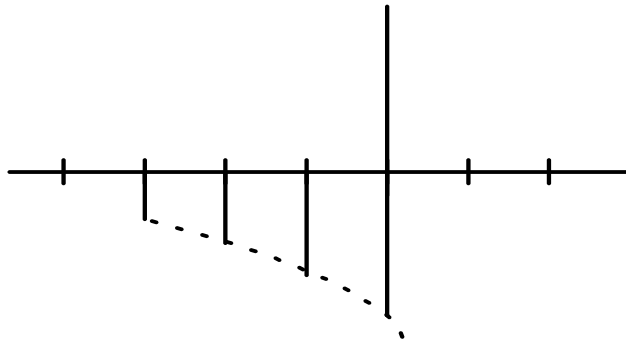
① Causal

② Unstable



Stability, Causality- illustration..(cont)

(4) $x[n] = -2^n u[-n-1]$



① Anti Causal

② Stable

$$X(z) = \sum_{n=-\infty}^{-1} -2^n z^{-n} = \frac{1}{1-2z^{-1}}$$

$$\text{RoC} : |z| < 2$$

① Inward

② $\mathbb{U} \subset \text{RoC}$

What do you find?

